Ordinary Differential Equation Solvers

MATH-151: Mathematical Algorithms in Matlab

October 16, 2023



Embracing Change

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  - Sometimes we know a lot about where we start and want to know what happens afterwards, this is an **initial value problem** or IVP
  - Other times we know what happens at two points and want to find what has happened in between, this is a **boundary value problem** or BVP

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- The first equation tells me how my acceleration changes my speed, then once I know my speed I can see how that changes my position!

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- What is the simplest way to do this? Since $y' = f(t, y)$ we can calculate $f(t, y_n)$ so we will use left rectangular rule to estimate the change!
- Doing this leaves us with the **forward Euler method**

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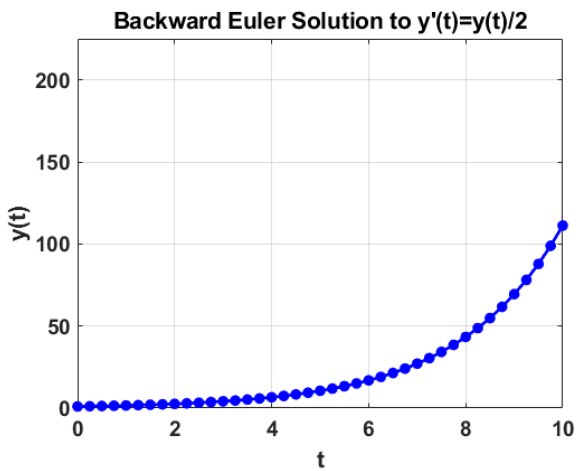
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- Let's try this with the IVP $y' = \frac{1}{2}y$ where $y(0) = 1$, with step size $h = 0.1$

$$y_{n+1} = y_n + h\frac{y_n}{2}$$

We can find the true solution is

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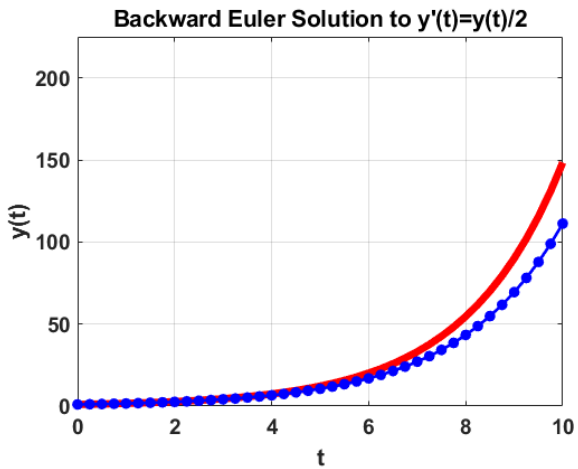
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Unfortunately, our approximation looks exponential, but it lags behind the true solution...



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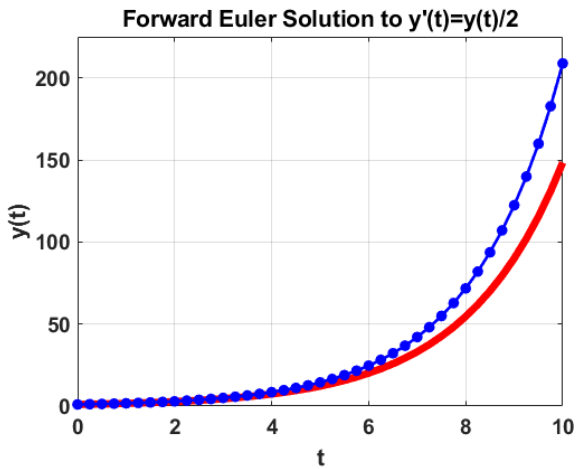
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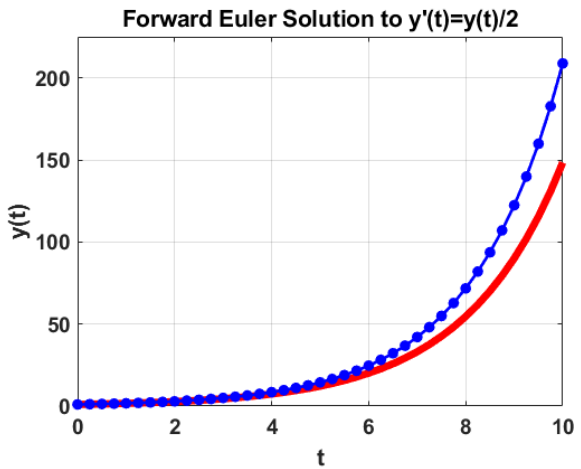


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- This still seems to be off, maybe we need to find a middle-ground between these two

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- The most commonly seen is the 4th-order Runge-Kutta algorithm, often called just RK4. It is similar to applying Simpson's rule to integrating our differential equation. We perform it as below

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$
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- A quick way to think about it is that we keep trying to take half-steps forward and refining our guess of our slope
- It looks like a lot of work at first, but once you get comfortable it works very well!

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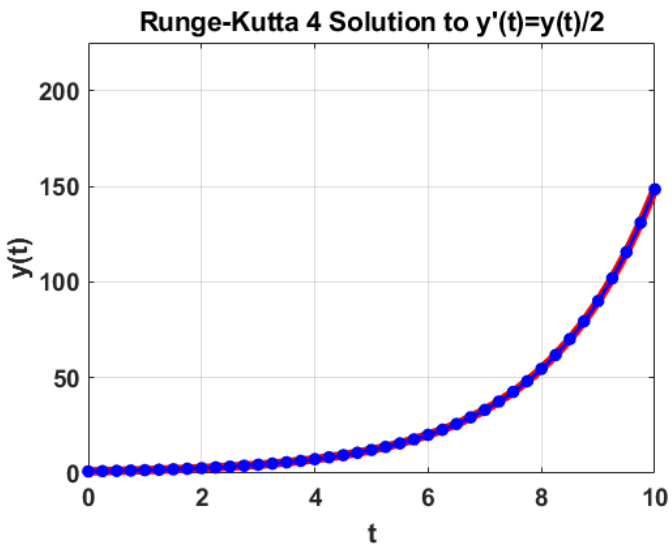
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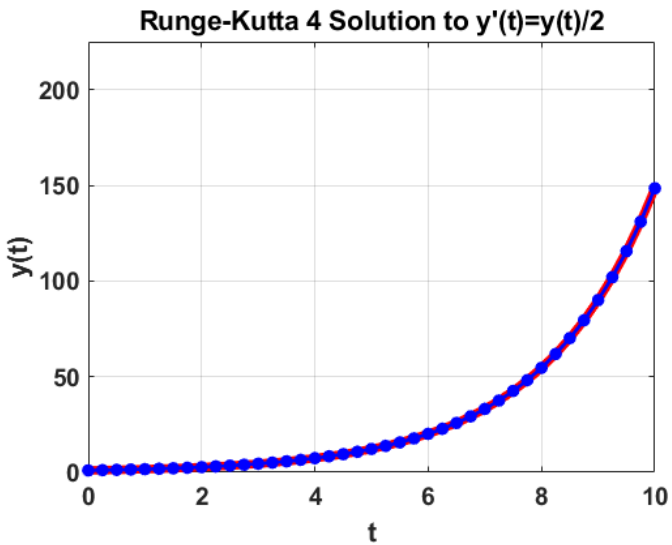
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- This is surprisingly straightforward to code up and it gives us VERY accurate solutions!
  - Our maximum error in this example was 0.0014

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- These are more difficult to solve, we don't even have enough information to start "marching forward" line we did for IVPs...

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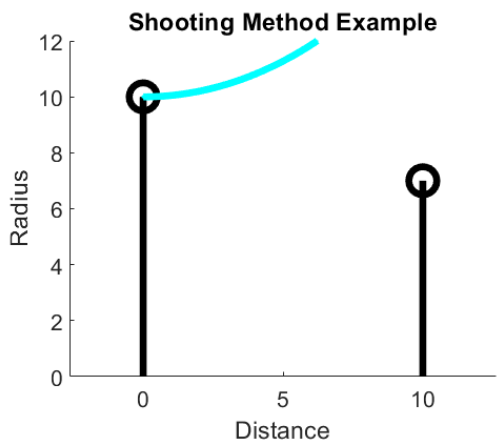
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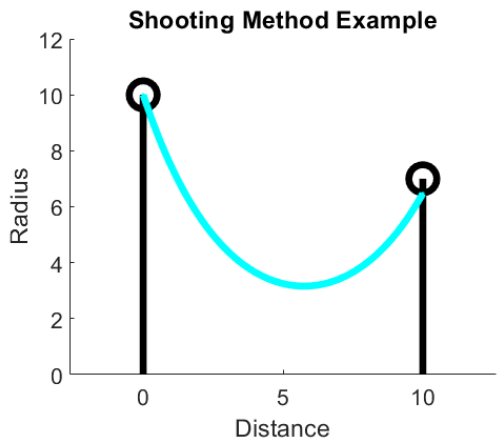


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Too high, lets aim lower



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We keep adjusting until

